Cavity Blockade: A way to make photons interact

Quantum Information with Atoms and Photons
Project Report

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1. Introduction

The Jaynes-Cummings (JC) model is the cornerstone of light-matter interaction, describing the coherent coupling between a two-level atom and a single quantized cavity mode. In the strong-coupling regime, the system enters the domain of quantum nonlinearity, where the presence of a single photon significantly shifts the energy required to add a second. This ***"photon blockade"*** transforms the cavity into a quantum gatekeeper, with modern applications ranging from the development of high-efficiency *quantum batteries* and microcavity devices to serving as a fundamental building block for deterministic single-photon sources in quantum communication networks.

The physical mechanism rests on the anharmonicity of the Jaynes–Cummings (JC) energy ladder. When an atom couples to a single cavity mode with strength $\tilde{g}$, the bare product states $|g, n\rangle$ (atom in ground state, n photons) and $|e, n-1\rangle$ (atom excited, $n-1$ photons) hybridise into dressed states (polaritons) $|n, \pm\rangle$ with energies $\pm\tilde{g}\sqrt{n}$. Because these energies scale as $\sqrt{n}$ rather than linearly in n, the one-photon and two-photon resonances are split by $\tilde{g}(\sqrt{2}-1) \approx 0.41\,\tilde{g}$. In the strong-coupling regime, where $\tilde{g}$ far exceeds the cavity linewidth κ and the atomic decay rate Γ, this anharmonic gap is spectrally resolved. A monochromatic drive tuned to the one-photon resonance therefore cannot simultaneously excite the two-photon transition, and the cavity is restricted to at most a single photon at any time.

The first experimental observation of this phenomenon was reported by Birnbaum *et al.* in *Nature* [1], using a single caesium atom trapped inside Fabry–Pérot optical cavity. The signature they measured was a sub-Poissonian photon-number distribution and, more directly, a two-photon correlation function $g^{(2)}(0) < 1$ at the one-photon polariton resonance .

In this report we reproduce the essential physics of that experiment through a theoretical and numerical study of the driven Jaynes Cummings model. The system is described by the Hamiltonian

$$\hat{H} = \omega_c \hat{a}^\dagger \hat{a} + \omega_a \left|e\right\rangle \left\langle e\right| + \tilde{g}\left(\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger\right) + \hat{H}_d, \tag{1}$$

where $\hat{H}_d = E\, e^{i\omega t}\hat{a} + \text{h.c.}$ is the coherent pumping field and $\hbar = 1$ throughout the report.

2. Physical Meaning of $\hat{H}_d$

Let us first understand the setup and interactions in the system and environment. The provided Hamiltonian is,

$$\hat{H} = \omega_c \hat{a}^\dagger \hat{a} + \omega_a \left|e\right\rangle \left\langle e\right| + \tilde{g}\left(\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger\right) + \hat{H}_d, \tag{2}$$

The first term $\omega_c \hat{a}^\dagger \hat{a}$ represents the cavity mode with the cavity resonance ω_c in which any photon can resonate and stay in cavity. Hence it is a single mode cavity. The second term $\omega_a \left|e\right\rangle \left\langle e\right|$ represents the stom with atomic transition frequency ω_a . Then we have the Jaynes Cummings atom-cavity coupling term followed by term that describes the external drive.

The driving term in the lab-frame Hamiltonian (2) is

$$\hat{H}_d = E\, e^{i\omega t}\hat{a} + E^* e^{-i\omega t}\hat{a}^\dagger, \tag{3}$$

where $E \in \mathbb{C}$ is the drive amplitude and ω is the laser frequency. Since E is not represented as an operator it is safe to assume it represents the classical nature, perhaps a consequence of large occupation of photons in coherent state. Hence, this term describes a classical, coherent monochromatic field, a laser injected into the cavity.

Physically, $E\, e^{i\omega t}\hat{a}$, presence of $\hat{a}$ denotes absorption of a cavity photon while supplying a quantum of energy ω from the classical field (stimulated absorption into the cavity), and its Hermitian conjugate $E^* e^{-i\omega t}\hat{a}^\dagger$ creates a cavity photon (emission back into the cavity mode). Together they describe exchange of energy between the external drive and cavity- Stimulated emission and absorption between cavity and the drive.

The amplitude $|E|$ controls the rate at which photons are injected; the phase of E sets the quadrature of the injected field. In the weak-drive limit $|E| \ll \kappa$ the mean intracavity photon number is small and the quantum nature of the field is most pronounced. Increasing $|E|$ beyond κ eventually overwhelms the blockade mechanism, driving the cavity into a more classical coherent state and risking optical bistability.

2.1 Physical Realization

Experimentally, this term is realized by sending a classical laser beam of frequency ω onto a partially transmitting mirror of the cavity. The external laser field at the mirror can be written as a classical wave

$E_{in}(t) = E\, e^{-i\omega t} + E^*\, e^{i\omega t}$

Through the input mirror, this field couples linearly to the internal cavity mode. Treating the external field as classical and the cavity mode as quantized, this coupling produces an effective interaction linear in the cavity operators, of the form $H_{int}(t) \propto Ee^{-i\omega t\hat{a}^\dagger} + E^* e^{i\omega t\hat{a}}$ After absorbing constants into a complex amplitude E and using the rotating-wave approximation to discard rapidly oscillating terms, one obtains the standard drive Hamiltonian

$$\hat{H}_d = E\, e^{i\omega t}\hat{a} + E^* e^{-i\omega t}\hat{a}^\dagger, \tag{4}$$

Thus, H_d is physically implemented by a coherent laser drive on the cavity, with E set

by the pump laser amplitude and cavity input coupling.

3. Hamiltonian in the Drive Frame and RWA

3.1 Rotating-frame unitary

To eliminate the explicit time dependence we move to the frame co-rotating with the drive at frequency ω via the unitary operator R

$$R = \exp\Big[i\omega t\Big(\hat{a}^\dagger\hat{a} + |e\rangle\langle e|\Big)\Big]. \tag{5}$$

The transformed Hamiltonian is

$$\hat{H}' = R\,\hat{H}\,R^\dagger + iR\dot{R}^\dagger. \tag{6}$$

Differentiating R with respect to time:

$$\dot{R}^\dagger = ie^{i\omega t(\hat{a}^\dagger\hat{a}+|e\rangle\langle e|)} - i\omega\Big(\hat{a}^\dagger\hat{a} + |e\rangle\langle e|\Big)\,e^{-i\omega t(\hat{a}^\dagger\hat{a}+|e\rangle\langle e|)}, \tag{7}$$

$$iR\dot{R}^\dagger = \omega\Big(\hat{a}^\dagger\hat{a} + |e\rangle\langle e|\Big). \tag{1}$$

3.2 Transformation of each Hamiltonian term

Term ② — Cavity. $RH_cR^\dagger = \omega\exp\Big[i\omega t\Big(\hat{a}^\dagger\hat{a} + |e\rangle\langle e|\Big)\Big]\,(\hat{a}^\dagger\hat{a})\exp\Big[i\omega t\Big(\hat{a}^\dagger\hat{a} + |e\rangle\langle e|\Big)\Big]$

Since $\Big[\hat{a}^\dagger\hat{a},\,\hat{a}^\dagger\hat{a},\,=\Big]\,0$ through the Baker–Campbell–Hausdorff series we obtain:

$$R\hat{H}_cR^\dagger = \omega_c\hat{a}^\dagger\hat{a}. \tag{2}$$

Term ③ — Atom. Similarly $[|e\rangle\langle e|\,,\,|e\rangle\langle e|\,,\,=]\,0$, hence

$$R\hat{H}_AR^\dagger = \omega_a\,|e\rangle\langle e|\,. \tag{3}$$

Term ④ — Atom–light coupling. Using $R\,\hat{a}\,R^\dagger = \hat{a}e^{-i\omega t}$, $R\,\hat{\sigma}_+\,R^\dagger = \hat{\sigma}_+e^{+i\omega t}$ and their conjugates, the phases cancel in each product:

$$\begin{aligned} R\hat{H}_{AL}R^\dagger &= \tilde{g}\Big(\hat{\sigma}_+e^{+i\omega t}\cdot\hat{a}e^{-i\omega t} + \hat{\sigma}_-e^{-i\omega t}\cdot\hat{a}^\dagger e^{+i\omega t}\Big) \\ &= \tilde{g}\Big(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^\dagger\Big). \end{aligned} \tag{4}$$

Term ⑤ — Drive.

$$\begin{aligned} R\hat{H}_d R^\dagger &= E\, e^{i\omega t}\cdot \hat{a}e^{-i\omega t} + E^* e^{-i\omega t}\cdot \hat{a}^\dagger e^{+i\omega t} \\ &= E\hat{a} + E^*\hat{a}^\dagger. \end{aligned} \tag{5}$$

The drive becomes *static* in the rotating frame; the driving Hamiltonian is eliminated.

3.3 Full rotating-frame Hamiltonian

Combining (1)–(5) and defining $\Delta_c \equiv \omega_c - \omega$, $\Delta_a \equiv \omega_a - \omega$:

$$\boxed{\hat{H}' = \Delta_c \hat{a}^\dagger\hat{a} + \Delta_a \left|e\right\rangle\left\langle e\right| + \tilde{g}\left(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^\dagger\right) + E\hat{a} + E^*\hat{a}^\dagger.} \tag{8}$$

3.4 Rotating-wave approximation

In the Hamiltonian (8) there are no time-dependent terms. The coupling $\tilde{g}(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^\dagger)$ contains only energy-conserving processes; the counter-rotating terms $\hat{\sigma}_+\hat{a}^\dagger$ and $\hat{\sigma}_-\hat{a}$ are absent because the original Hamiltonian (2) was already written in the RWA. Consequently there is nothing to drop and no additional rotating-wave approximation is required.

4. Steady-State Integration

4.1 Lindblad master equation

The cavity and atom are open systems: the cavity loses photons through its mirrors at rate κ, and the atom decays spontaneously at rate Γ. Hence we need to modify the Lindbald Master equation under Born–Markov approximation with Lindbald dissipators, with the Lindblad dissipator

$$\mathcal{D}\left[\hat{L}\right]\hat{\rho} \equiv \hat{L}\hat{\rho}\hat{L}^\dagger - \tfrac{1}{2}\hat{L}^\dagger\hat{L}\hat{\rho} - \tfrac{1}{2}\hat{\rho}\hat{L}^\dagger\hat{L}. \tag{9}$$

we obtain the Lindblad master equation

$$\boxed{\frac{d\hat{\rho}}{dt} = -i\left[\hat{H}',\, \hat{\rho}\right] + \kappa\,\mathcal{D}[\hat{a}]\,\hat{\rho} + \Gamma\,\mathcal{D}[\hat{\sigma}_-]\,\hat{\rho},} \tag{10}$$

4.2 Resonance case and parameter choice

At resonance ($\omega = \omega_a = \omega_c$) with $\tilde{g} = 0$, the Hamiltonian reduces to $\hat{H}' = E\hat{a} + E^*\hat{a}^\dagger$. We choose $E = 1.0$, $\kappa = 0.5$, giving $\langle\hat{n}\rangle_{\rm ss} = 16$. After many iteration it is found that Fock-space at $N = 50$ is safely adequate. We set $\Gamma = 0.1$ and integrate from $|g, 0\rangle$ to $t_{\max} = 60 \gg 1/\kappa$ using QuTiP [2].

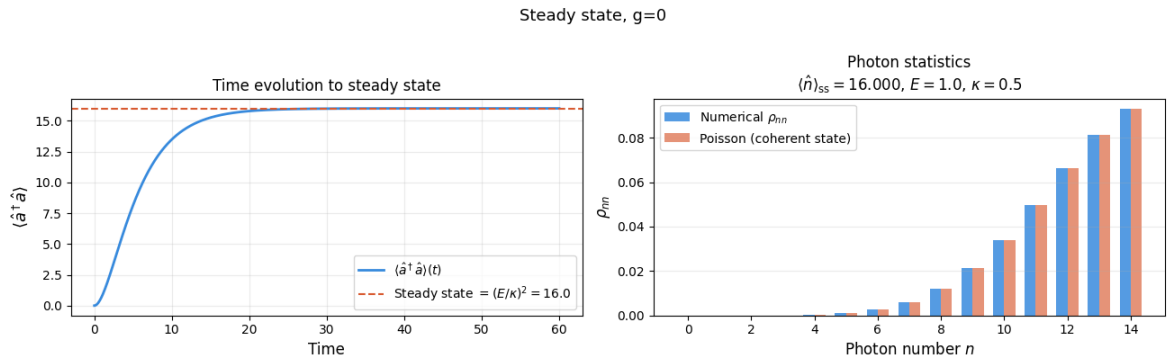


Figure 1: Steady State integration

4.3 Photon number distribution

At steady state the cavity is in a coherent state with $|\alpha_{\rm ss}|^2 = 4E^2/\kappa^2$ and a Poissonian photon distribution:

$$\rho_{nn} = e^{-\langle\hat{n}\rangle_{\rm ss}} \frac{\langle\hat{n}\rangle_{\rm ss}^n}{n!}. \tag{11}$$

Inference: The plots clearly show the numerical simulation and analytical distribution overlaps at 16 verifying the choices of operators and number of photon states and other parameters are correct.

5. Steady State Under Strong Coupling and Weak Drive

In the uncoupled case ($\tilde{g} = 0$) we set $\omega = \omega_c = \omega_a$ and found a coherent steady state. In the presence of a finite coupling $\tilde{g} \neq 0$, however, the bare cavity and atom frequencies are no longer the relevant resonances. The atom–cavity coupling hybridises the bare states into the dressed-state (polariton) doublets of Eq. (18), shifting the one-photon resonance to

$$\omega_{1,\pm} = \omega_c \pm \tilde{g}, \tag{12}$$

so that a drive at $\omega = \omega_c$ is now detuned from both polariton branches by $\pm\tilde{g}$ and the previous resonance condition is no longer valid.

5.1 New resonance condition: lower one-photon polariton

To maximise the coupling between the drive and the intracavity field we tune ω to resonance with the lower one-photon dressed state $|1,-\rangle$:

$$\omega = \omega_c - \tilde{g} \quad \Longrightarrow \quad \Delta_c = \Delta_a = \tilde{g}. \tag{13}$$

With this choice the rotating-frame Hamiltonian (8) becomes

$$\hat{H}' = \tilde{g}\,\hat{a}^\dagger\hat{a} + \tilde{g}\,|e\rangle\langle e| + \tilde{g}\left(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^\dagger\right) + E\hat{a} + E^*\hat{a}^\dagger, \tag{14}$$

and the master equation (10) is integrated numerically to find the new steady state.

To verify the steady state we used QuTip to find the population of photons in steady state: Steady State Populations for E=0.05: P(0 photons, ground state): 0.679802 : Since E is weak, sys is mostly in $|g,0\rangle$ P(1 photon, ground state): 0.160062 :Due to resonance with lower polariton P(0 photons, excited state): 0.160000 :Due to resonance with lower polariton P(Total > 1 photon): 1.358816e-04 :Blockade, almost 0 population

5.2 Strong-coupling regime

We operate in the strong-coupling condition

$$\tilde{g} \gg \Gamma,\, \kappa, \tag{15}$$

with numerical values $\tilde{g} = 5$, $\kappa = 0.1$, $\Gamma = 0.05$, giving $\tilde{g}/\kappa = 50 \gg 1$. In this regime the anharmonic gap between the one-photon and two-photon polariton resonances,

$$\delta_{\text{block}} = \tilde{g}\big(\sqrt{2} - 1\big) \approx 0.41\, \tilde{g}, \tag{16}$$

is much larger than the cavity linewidth $\kappa/2$, so the dressed states are spectrally well resolved. The steady-state photon distribution becomes strongly sub-Poissonian, as quantified by the Mandel Q parameter

$$Q = \frac{\langle \hat{n}^2 \rangle - \langle \hat{n} \rangle^2}{\langle \hat{n} \rangle} - 1 < 0, \tag{17}$$

in contrast to $Q = 0$ for the coherent state found in the uncoupled case.

5.3 Role of the pumping strength E

By refering to the output from QuTip analysis we conclude that the drive amplitude E plays a critical role in determining whether the blockade is active and whether the system remains in a stable steady state. Three distinct regimes can be identified:

Weak drive: $E \ll \kappa$. The drive is too weak to populate states beyond $n = 1$ appreciably. The cavity occupies a superposition dominated by $|0\rangle$ and $|1\rangle$, with $g^{(2)}(0) \ll 1$ and $Q < 0$. This is the true photon-blockade regime: at most one photon resides in the cavity at any time.

Intermediate drive: $E \sim \kappa$. The drive is now strong enough to partially overcome the anharmonic gap. The photon distribution broadens, $Q \to 0$ and $g^{(2)}(0) \to 1$: the blockade is progressively washed out and the statistics approach Poissonian.

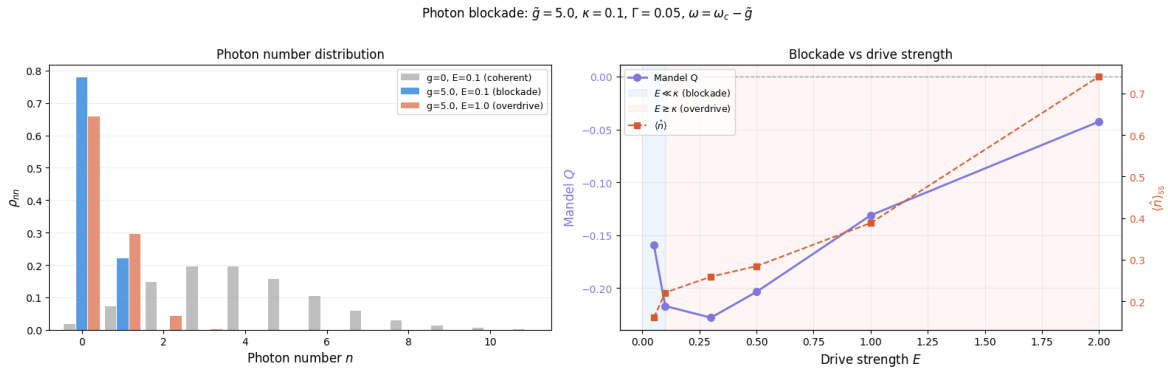


Figure 2: Photon Blockade

Strong drive: $E \gtrsim \kappa$. When E is comparable to or exceeds κ, the drive forces significant occupation of $n \geq 2$ states despite the anharmonic detuning. The system can exhibit super-Poissonian statistics ($Q > 0$, $g^{(2)}(0) > 1$), and for sufficiently large E the mean-field equations become bistable: two stable steady states coexist for the same set of parameters, and the numerical steady-state solver may return different solutions depending on the initial condition. Care must therefore be taken to keep E in the weak-drive regime when studying the blockade.

6. Possibility of Two-Photon States

6.1 Dressed states

With $\tilde{g} \neq 0$ the eigenstates of the JC Hamiltonian are the dressed states (polaritons):

$$|n, \pm\rangle = \frac{1}{\sqrt{2}}(|g, n\rangle \pm |e, n-1\rangle)\,, \qquad E_{n,\pm} = \pm\tilde{g}\sqrt{n}. \tag{18}$$

6.2 Anharmonicity and blockade gap

The two-photon dressed states lie at $\pm\tilde{g}\sqrt{2}$, not at $\pm 2\tilde{g}$. A drive resonant with $|1, -\rangle$ is therefore detuned from the two-photon transition by

$$\delta_{\text{block}} = \tilde{g}\left(\sqrt{2} - 1\right) \approx 0.41\,\tilde{g}. \tag{19}$$

For $\tilde{g} \gg \kappa$, this gap exceeds the cavity linewidth and the two-photon state $|2, -\rangle$ cannot be populated: the photon blockade is active.

6.3 Drive resonance condition

Driving resonantly into $|1, -\rangle$ requires

$$\omega = \omega_c - \tilde{g} \quad \Longrightarrow \quad \Delta_c = \Delta_a = \tilde{g}. \tag{20}$$

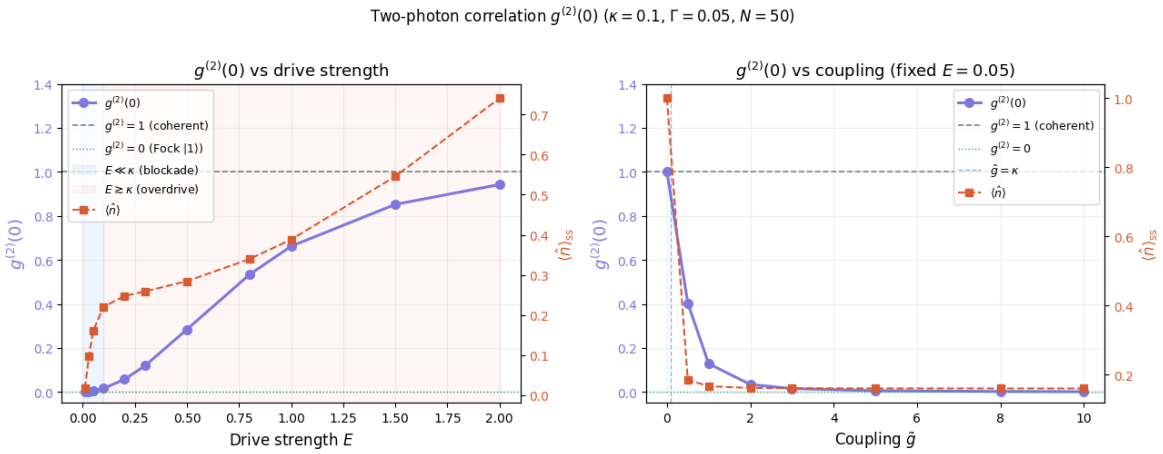


Figure 3: g^2*correlation*

The rotating-frame Hamiltonian becomes

$$\hat{H}' = \tilde{g}\,\hat{a}^\dagger\hat{a} + \tilde{g}\,|e\rangle\langle e| + \tilde{g}\left(\hat{\sigma}_+\hat{a} + \hat{\sigma}_-\hat{a}^\dagger\right) + E\hat{a} + E^*\hat{a}^\dagger. \tag{21}$$

7. Two-Photon Correlation Function $g^{(2)}(0)$

The two-photon correlation function is given as:

$$g^{(2)}(0) = \frac{\langle \hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a}\rangle}{\langle \hat{a}^\dagger\hat{a}\rangle^2} = \frac{\langle \hat{n}(\hat{n}-1)\rangle}{\langle \hat{n}\rangle^2}. \tag{22}$$

It compares the probability of detecting two simultaneous photons with the probability expected if photons arrived as independent, classical random events.

7.1 Interpretation

- $g^{(2)}(0) = 1$: Poissonian: photons arrive independently (coherent state; no blockade).
- $g^{(2)}(0) < 1$: sub-Poissonian antibunching; photon blockade is active.
- $g^{(2)}(0) = 0$: perfect single-photon Fock state $|1\rangle$.
- $g^{(2)}(0) > 1$: super-Poissonian bunching; two-photon resonance or bistability.

7.2 Numerical results: fig.3

With $\tilde{g} = 5$, $\kappa = 0.1$, $\Gamma = 0.05$, $E = 0.05$ (strong coupling, weak drive), the numerical solution gives $g^{(2)}(0) \ll 1$, confirming antibunching. Increasing E toward κ destroys the blockade and $g^{(2)} \to 1$. For $E \gtrsim \kappa$ super-Poissonian fluctuations emerge.

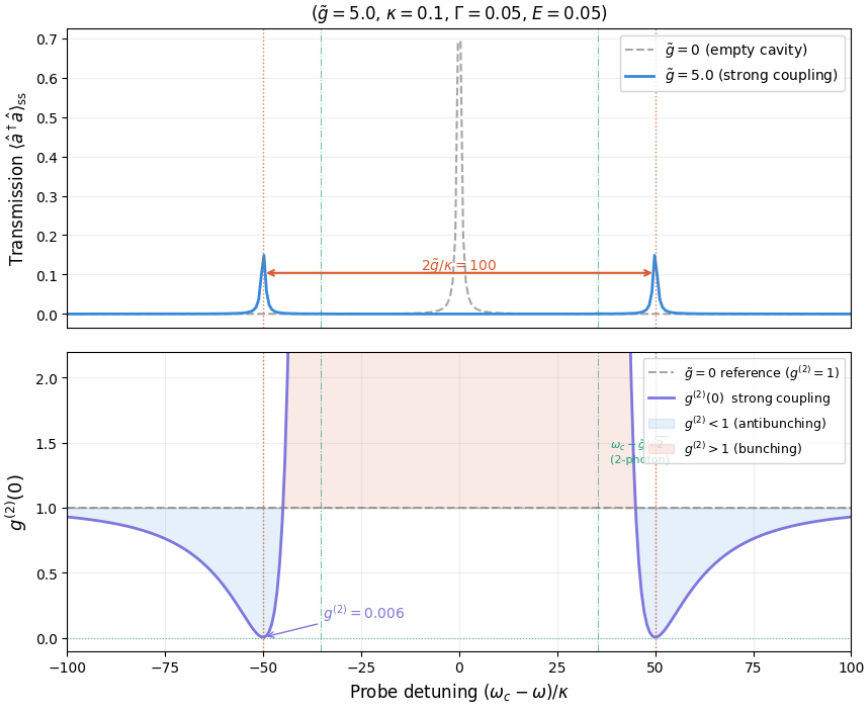


Figure 4: ω scan

8. Scanning ω Around Resonance

8.1 comparison with Birnbaum *et al.*

The scan fig.4 reproduces the key features of Fig. 2a of Birnbaum *et al.* [1]:

- **Polariton transmission peaks** at $\Delta = \pm\tilde{g}$ (vacuum Rabi splitting $2\tilde{g}/\kappa$ linewidths wide) The splitting $2\tilde{g} = 100\kappa$ confirms the system is in strong coupling regime, where the atom and cavity occurs much faster than the dissipative rates.
- **Antibunching** $g^{(2)}(0) < 1$ at the polariton peaks –the spectroscopic signature of the photon blockade.
- **Bunching** $g^{(2)}(0) > 1$ near $\Delta = \pm\tilde{g}/\sqrt{2}$ — the two-photon resonance. This indicates Two=photon Tunneling, at specific detunings the laser is two-photon resonant with $|2, \pm\rangle$ states, bypassing the suppressed single-photon manifold.
- $g^{(2)}(0) \approx 1$ away from all resonances.

This pattern is the clearest spectroscopic fingerprint of the anharmonic Jaynes–Cummings ladder and directly demonstrates photon–photon interaction mediated by the atom.

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